Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2024-P2-Model)** **Date: 26-04-2026**

Time: 02.30Pm to 05.30Pm **GTA-27** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	**B**	2	**C**	3	**D**	4	**D**	5	**ABCD**	6	**ABCD**
7	**AC**	8	**321**	9	**24**	10	**14**	11	**4**	12	**3**
13	**61**	14	**0**	15	**14**	16	**1488**	17	**42**		

PHYSICS

18	**B**	19	**B**	20	**C**	21	**A**	22	**AC**	23	**AC**
24	**ABC**	25	**2**	26	**5**	27	**6**	28	**6**	29	**5**
30	**7**	31	**60**	32	**2.9 to 3.1**	33	**332 to 335**	34	**315 to 318**		

CHEMISTRY

35	**A**	36	**C**	37	**C**	38	**A**	39	**B**	40	**AC**
41	**ABD**	42	**6**	43	**9**	44	**5**	45	**5**	46	**9**
47	**407**	48	**0.96**	49	**6**	50	**32**	51	**7**		

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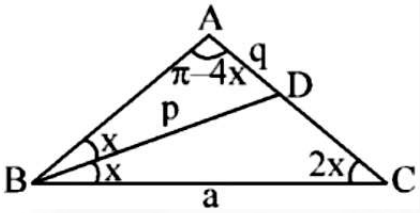
SOLUTIONS
MATHEMATICS

1. $BC = BD + AD$

$a = p + q$ or $\frac{a}{p} = 1 + \frac{q}{p}$

Using Sine law in ΔBDC and in ΔABD

$$\frac{\sin 3x}{\sin 2x} = 1 + \frac{\sin x}{\sin 4x} \Rightarrow \frac{\sin 3x \cdot \sin 4x - \sin x \cdot \sin 2x}{\sin 2x \cdot \sin 4x} = 1$$



$$2\sin 3x \cdot \sin 4x - 2\sin x \cdot \sin 2x = 2\sin 2x \cdot \sin 4x$$

$$(\cos x - \cos 7x) - (\cos x - \cos 3x) = \cos 2x - \cos 6x$$

$$\cos x - \cos 7x = \cos 2x - \cos 6x$$

$2\sin 5x \cdot \sin 2x = 2\sin 4x \cdot \sin 2x$ as $\sin 2x \neq 0$

Hence $\sin 5x = \sin 4x$

$5x + 4x = 180^\circ$

$\Rightarrow x = 20^\circ$; hence $\angle A = 180^\circ - 80^\circ = 100^\circ$

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2. Put $x = 6n + r \qquad r \in [0,6)$

$$6n + r = \left[\frac{6n+r}{2}\right] + \left[\frac{12n+2r}{3}\right] - \left[\frac{6n+r}{6}\right]$$

$$= 3n + \left[\frac{r}{2}\right] + 4n + \left[\frac{2r}{3}\right] - n - \left[\frac{r}{6}\right]$$

$$\Rightarrow r = \left[\frac{r}{2}\right] + \left[\frac{2r}{3}\right]$$

r satisfy 0, 2, 3, 4, 5 but not 1.

3. $$\ell nS = \ell n\left(\prod_{i=1}^{2n}\left(n^2 + i^2\right)^{1/n}\right) - 4 \cdot \ell nn$$

$$= \frac{1}{n}\sum_{i=1}^{2n} \ell n\left(n^2 + i^2\right) - 4 \cdot \ln n$$

$$= \frac{1}{n} \cdot \sum 2\ell nn + \ell n\left(1 + \frac{i^2}{n^2}\right) - 4 \cdot \ln n$$

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$$= \int_0^2 \ell n\left(1+x^2\right)\cdot dx = 2\ell n5 - 4 + 2\tan^{-1}2$$

$$\Rightarrow S = 25\cdot e^{2\tan^{-1}2-4}$$

4. Conceptual

5. $f'(x) > 0 \forall x \in (1,\infty)$ so $f(x)$ is strictly increasing.

So, $f(x) \geq f(1) \forall x \in (1,\infty) \Rightarrow f'(x) \leq \dfrac{1}{x^2+1}$

So, $f(x) = 1 + \int_1^x f'(t)\,dt \leq 1 + \int_1^x \dfrac{1}{t^2+1}dt \leq 1 + \left|\tan^{-1}t\right|_1^x \leq 1 + \tan^{-1}x - \dfrac{\pi}{4}$

$x \to \infty, f(x) \to 1 + \dfrac{\pi}{4}$

6. **(a)** Given $P(A \cap B) = P(A) + P(B)$ ..... (1)

$$P\left(A \cap (B \cup C)\right) = P(A)P(B \cup C) \quad \text{..... (2)}$$

$$P\left(A \cap (B \cap C)\right) = P(A)P(B \cap C) \quad \text{..... (3)}$$

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To prove that $P(A \cap C) = P(A)P(C)$

From Eqn. (2), we get

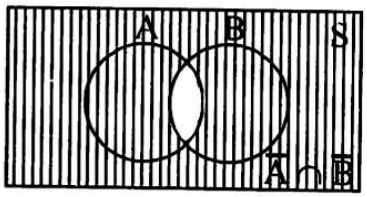
$$P\left[(A \cap B) \cup (A \cap C)\right] = P(A)\left[P(B) + P(C)\ \ P(B \cap C)\right]$$

$$\Rightarrow \quad P(A \cap B) + (A \cap C) \quad P(A \cap B \cap C)$$

$$= P(A)P(B) + P(A)P(C) \quad P(A)P(B \cap C)$$

$\Rightarrow \quad P(A \cap C) = P(A)P(C) \Rightarrow$ (a) is correct.

(b) See by Venn diagram



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(c) If nine letters are in the correct envelopes, the tenth must be also in the correct envelope, so the probability is zero.

(d) Let $P(S)$ = probability of successful flight

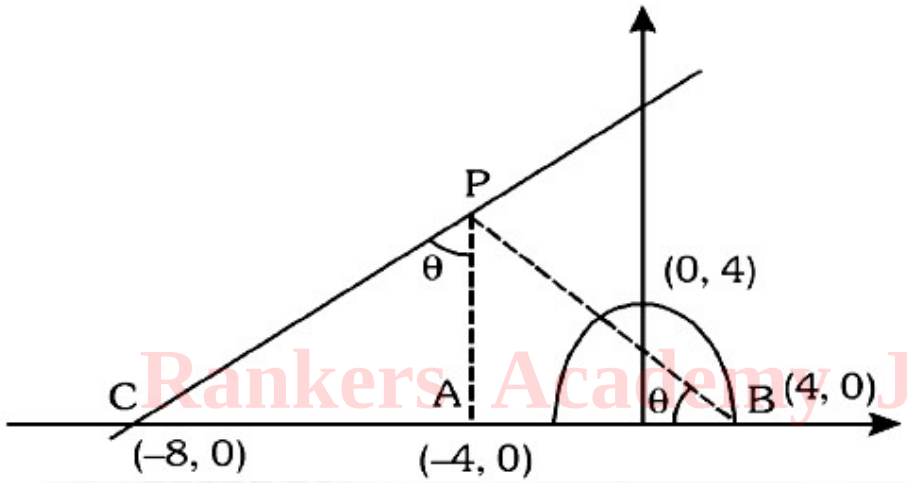
Let $P(\bar{S})$ = probability of unsuccessful flight

$$P(S)=1 \quad P(\bar{S})=1-(0.01)^3=1-0.000001=0.999999$$

7. $\angle APB$ is maximum if L is the tangent to the circumcircle of ΔPAB

$\therefore CP^2=CA\cdot CB=4\times 12=48$

$\Delta CPA\sim\Delta CBP$



$$\frac{CP}{CB}=\frac{PA}{PB}=\frac{\sqrt{48}}{12}=\frac{1}{\sqrt{3}}$$

$PB=\sqrt{3}PA$

8. $$\prod_{k=0}^{12}\left(2-2\omega^k+\omega^{2k}\right)=\prod_{k=0}^{12}\left(\omega^k-(1+i)\left(\omega^k-(1-i)\right)\right)=\prod_{k=0}^{12}\left((1+i)-\omega^k\right)\left((1-i)-\omega^k\right)$$

We consider the polynomial $x^{13}-1=0$ whose roots are the 13th root of unity.

Let $\omega^0,\omega^1,\omega^2,.....\omega^{12}$ are the roots of this equation.

$$\Rightarrow x^{13}-1=\left(x-\omega^0\right)\left(x-\omega^1\right)\left(x-\omega^2\right)\left(x-\omega^3\right).....\left(x-\omega^{12}\right)$$

Put $x=1+i$ and $x=1-i$

$$\left((1+i)^{13}-1\right)\left((1-i)^{13}-1\right)=(i)(-i)\prod_{k=0}^{12}\left((1+i)-\omega^k\right)\left((1-i)-\omega^k\right)$$

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$$\Rightarrow \prod_{k=0}^{12}\left((1+i)-\omega^k\right)\left((1-i)-\omega^k\right)=\left(\left(\sqrt{2}e^{\frac{\pi i}{4}}\right)^{13}-1\right)\left(\left(\sqrt{2}e^{\frac{-\pi i}{4}}\right)^{13}-1\right)$$

The given expression $=(2)^{13}-\left(\sqrt{2}\right)^{13}\left[e^{\frac{-13i\pi}{4}}+e^{\frac{13i\pi}{4}}\right]+1$

$$=2^{13}-2^{\frac{15}{2}}\left(-\frac{1}{\sqrt{2}}\right)+1=2^{13}+2^7+1=8321$$

9. Substitute $x=\tan\theta$ to write this as $-2\int_0^{\pi/4}\left(\cos^2\theta-\sin^2\theta\right)^2\ell og(\cos\theta)d\theta$.

Now, $\left(\cos^2\theta-\sin^2\theta\right)^2=\cos^2(2\theta)=\frac{1+\cos(4\theta)}{2}$ so this integral becomes

$$-\int_0^{\pi/4}\ell og(\cos\theta)d\theta-\int_0^{\pi/4}\cos(4\theta)\ell og(\cos\theta)d\theta$$

We evaluate the second integral by parts to get

$$\int_0^{\pi/4}\cos(4\theta)\ell og(\cos\theta)d\theta=\frac{1}{4}\sin(4\theta)\ell og(\cos\theta)\Big|_{\theta=0}^{\pi/4}+\frac{1}{4}\int_0^{\pi/4}\sin(4\theta)\tan\theta d\theta$$

$$=\frac{1}{2}\int_0^{\pi/4}\frac{\sin(2\theta)\cos(2\theta)\sin\theta}{cis\theta}d\theta$$

$$=\int_0^{\pi/4}\sin^2(\theta)\cos(2\theta)d\theta$$

$$=\frac{1}{2}\int_0^{\pi/4}[1-\cos(2\theta)]\cos(2\theta)d\theta$$

$$=\frac{1}{4}\int_0^{\pi/4}\left(\cos\theta-\cos^2\theta\right)d\theta$$

$$=\frac{1}{4}\left(1-\frac{\pi}{4}\right)$$

$$=\frac{\pi-4}{16}$$

Thus the integral we're trying to find is

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$$=\frac{\pi-4}{16}-\int_0^{\pi/4}\ell og\cos\theta\cdot d\theta$$

Let $C=\int_0^{\pi/4}\ell og(\cos\theta)\cdot d\theta$ and $S=\int_0^{\pi/4}\ell og\sin\theta d\theta$

$$C+S=\int_0^{\pi/4}\ell og\left(\frac{\sin 2\theta}{2}\right)\cdot d\theta=\frac{-\pi}{4}\ell og2+\frac{1}{2}\int_0^{\pi/2}\ell og\sin\theta d\theta$$

$$=-\frac{\pi}{4}\ell og2+\frac{1}{2}\cdot\left(-\frac{\pi}{2}\ell og2\right)=-\frac{\pi}{2}\ell og2$$

$$C-S=\int_0^{\pi/4}\ell og(\cot\theta)\cdot d\theta=-\int_0^{\pi/4}\ell og(\tan\theta)d\theta$$

$$=-\int_0^1\frac{\ell ogt}{1+t^2}\cdot dt=-I$$

$$\Rightarrow C=-\left(\frac{\pi\ell og2+2I}{4}\right)$$

Required integral $=\frac{\pi-4}{16}+\frac{\pi\ell og2+2I}{4}$

$$=\frac{\pi-4+4\pi\cdot\ell og2+8.I}{16}$$

10. Put $x=r\cos\theta, y=r\sin\theta$

$$r^3\cdot(1+\sin\theta\cdot\cos\theta)=88$$

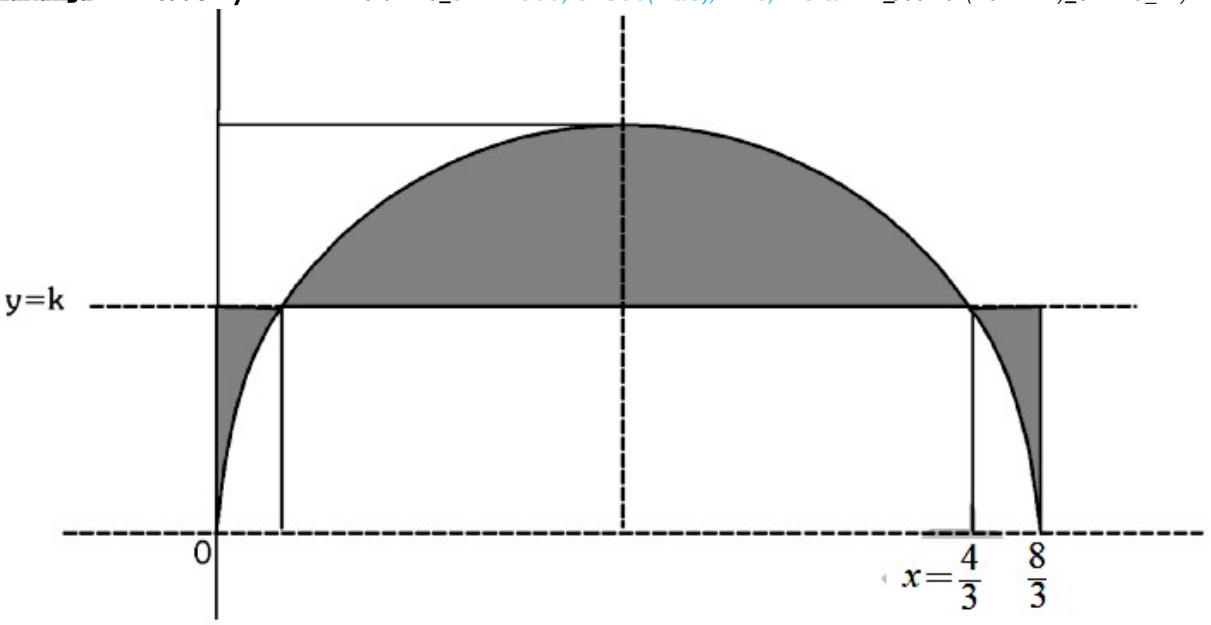
$$r^3\cdot(1-\sin\theta\cdot\cos\theta)=40$$

$$\Rightarrow\frac{1+SC}{1-SC}=\frac{11}{5}\Rightarrow\sin\theta\cdot\cos\theta=\frac{3}{8}$$

$$\Rightarrow\sin 2\theta=\frac{3}{4}$$

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11.



Given integral equals total shaded area, which equals

$g(K)=2\int_0^K f^{-1}(x)+2\int_K^{16/3}\frac{4}{3}-f^{-1}(x)$ which is min. when $g'(K)=0$.

$\Rightarrow g'(K)=f^{-1}(K)-f^{-1}\left(\frac{16}{3}\right)+f^{-1}(K)=0$

$\Rightarrow f^{-1}(K)=\frac{1}{2}\cdot f^{-1}\left(\frac{16}{3}\right)=\frac{2}{3}$

$\Rightarrow K=f\left(\frac{2}{3}\right)=4$

12. A and B are (1, 2, 3) and (2, 1, 2) respectively.

If $AC=d_1$ and $BC=d_2$, then volume of tetrahedron $=\frac{1}{6}\cdot ABd_1d_2=\frac{\sqrt{3}}{6}d_1d_2$.

Also, $CD^2=d_1^2+d_2^2+3\Rightarrow d_1^2+d_2^2=24$

Using AM, GM $=\frac{d_1^2+d_2^2}{2}\geq d_1d_2\Rightarrow d_1d_2\leq 12\Rightarrow V=2\sqrt{3}$

13. $\sum_{r=0}^{10}\frac{{}^{10}C_r}{{}^{30}C_{10+r}}=\frac{1}{{}^{30}C_{20}\cdot{}^{20}C_{10}}\sum_{r=0}^{10}{}^{10+r}C_r\cdot{}^{20-r}C_{10}$

${}^{10+r}C_r=$ Coefficient of x^r in $(1-x)^{-11}$

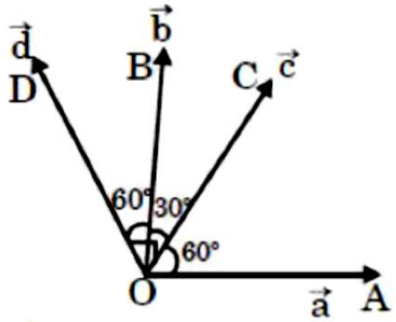
${}^{20-r}C_{10-r}=$ Coefficient of x^{10-r} in $(1-x)^{-11}$

$\Rightarrow\sum{}^{10+r}C_r\cdot{}^{20-r}C_r=$ Sum of coefficient of x^{10} in $(1-x)^{-22}$

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$= {}^{31}C_{10}$

14&15



($\hat{p}$ is unit vector $\perp$ to the plane)

$\sin\theta_1 \sin\theta_2 \cos\phi = 1, \cos\alpha = \frac{1}{2}$

$\theta_1 = \theta_2 = 90^\circ, \phi = 0, \alpha = \frac{\pi}{3}$

$\Rightarrow OABCD$ are coplanar

$\Delta ABC = \frac{1}{2}\left|\vec{a}\times\vec{b} + \vec{b}\times\vec{c} + \vec{c}\times\vec{a}\right|$

$= \frac{1}{2}\left|2\hat{p} - 3\hat{p} - \frac{3\sqrt{3}}{2}\hat{p}\right| = \frac{1}{2}\left|1 + \frac{3\sqrt{3}}{2}\right|$

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16&17

$|adjA| = 576 \Rightarrow |A| = 24$ or -24

Here $f(x) = px + q$ (linear function)

$f(0) = q = |A| = 24$

$p = f'(0) = \Sigma\alpha\beta - (a+b)(\Sigma\alpha) + a^2 + b^2 + ab$

= Sum of elements of adj A

= 18

$\Rightarrow f(x) = 18x + 24$

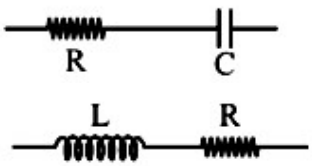
$GE = f(30) + f(50) = 18(30+50) = 24 + 24 = 1440 + 48 = 1488$

Consider $y = f(x) = px + q$

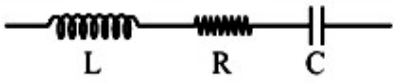
$y = g(x) = -(x-\alpha)(x-\beta)(x-\gamma)$

PHYSICS

18.



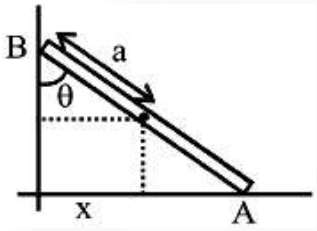
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$$P=\frac{V^2}{Z^2}R$$

$$P_1=\frac{V^2}{\sqrt{X_C^2+R^2}}R, \qquad P_2=\frac{V^2R}{\sqrt{X_L^2+R^2}}$$

$P_3=\frac{V^2R}{\sqrt{O+R^2}}$. Given $X_L=X_C$. Clearly $P_1=P_2<P_3$

19. 

$$\frac{d\theta}{dt}=2\Rightarrow\theta=2t$$

$$x=a\sin\theta$$

$$\left[x=a\sin(2t)\right]$$

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20. Relation between depth and height of water is $d=(H-x)+\frac{x}{\mu}$

21. Maximum angle which velocity vector at any instant can make with initial direction of projection is $80^\circ<90^\circ$. So distance of projectile from thrower always increases. Angle between $\vec{a}$ and $\vec{v}$ lies between 130° and 50°.

22. By the symmetry, the field must be directed along the axis at all points inside the cylinder.
For any Gaussian box drawn inside, the charge enclosed and therefore the flux must be zero. So the field must be uniform.

23. From figure $\phi=10\sin\omega t$,

When $\omega=\frac{2\pi}{4}=\frac{\pi}{2}$

$\therefore\phi=10\sin\frac{\pi}{2}t;$

Now $i=\frac{e}{R}=\frac{1}{R}\left|\frac{d\phi}{dt}\right|=\frac{1}{R}\times10\cos\frac{\pi t}{2}\times\frac{\pi}{2}$

$=\frac{5\pi}{2}\cos\frac{\pi t}{2}$

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$\Rightarrow \frac{d\phi}{dt}$ is zero (from fig.) at t = 1, 3, 5 sec

$\therefore$ i = zero at t = 1, 3, 5 sec

Total charge $Q = \int_0^b idt = \int_0^6 \frac{5\pi}{R}\cos\frac{\pi t}{2} = 0$

24. $Mg = kx$

$M = \frac{6000\times 0.1}{10} = 60kg$

$a_{\max} = A\omega^2 \le g$ (for not loosing contact)

$A = \frac{10}{100} = 0.1m$

25. $\lambda_{th} = \frac{hc}{eV_a}$

$\frac{1}{\lambda_{K\alpha}} = R(z-1)^2\left(\frac{1}{1^2}-\frac{1}{2^2}\right)$

$\frac{13}{10}(\lambda_{K\alpha}-\lambda_{th}) = \left(\lambda_{K\alpha}-\frac{\lambda_{th}}{2}\right)$

$\frac{3}{10}\lambda_{K\alpha} = \left(\frac{13}{10}-\frac{1}{2}\right)\lambda_{th}$

$\frac{3}{10}\left(\frac{4\times 10^{-7}}{3(z-1)^2}\right) = \left(\frac{8}{10}\right)\frac{12.4\times 10^{-7}}{15.5\times 10^3}$

$\Rightarrow \frac{5000}{8} = (z-1)^2$

$625 = (z-1)^2 \Rightarrow z = 26$

26. For $v_{\min}$ to reach the origin it must cross the point $x = 2L$.

$\therefore$ By work energytheorem

$\Delta KE = W$ (or area)

$0 - \frac{1}{2}mv^2 = -\frac{1}{2}\times L\times E$

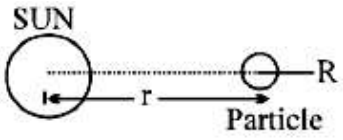
$v = \sqrt{\frac{LE_0}{m}}$ $\qquad v = 5m/s$

27. $f\lambda = v = \frac{10\lambda v}{d} \Rightarrow \lambda = \frac{d}{10}$

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Maximum path diff. can be $\frac{d}{3}$. Hence, number of maximum will be 6.

28.



Energy incident/sec on the particle

$$=\frac{P}{4\pi r^2}\times\pi R^2$$

$$=\frac{PR^2}{4r^2}$$

$\vec{F}$ due to striking of photon

$\vec{F}_1=\frac{\vec{dp}}{dt}$ = Change in lin. mom. of 1 photon in collision $\times$ No. of photons striking per sec.

$$=\frac{h\nu}{C}\times\frac{PR^2}{4r^2\times h\nu}$$

$$=\frac{PR^2}{4r^2C}$$

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Gravitational force on the particle

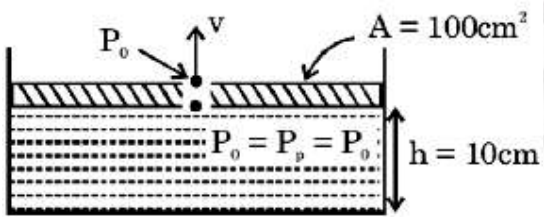
$$\vec{F}_2=\frac{GM_Sm}{r^2}=\frac{GM_S}{r^2}\times\frac{4}{3}\pi R^3\rho$$

$$\left|\vec{F}_1\right|=\left|-\vec{F}_2\right|$$

$$\frac{PR^2}{4r^2C}=\frac{GM_s\times 4\pi R^3\rho}{3r^2}$$

$$\Rightarrow R=\frac{3P}{16GM_S\pi\rho C}=0.6$$

29.



Weight of piston = mg

Pressure due to piston

$$P_p=\frac{mg}{A}=10^5 \text{ pascal.}$$

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Applying Bernoulli's equation just inside and just outside the hole.

$$P_p + P_0 = P_0 + \frac{1}{2}\rho v^2$$

$$10^5 = \frac{1}{2}\rho v^2 \Rightarrow v = 10\sqrt{2}\, m/s$$

Rate of volume flow

$\frac{dV}{dt} = av$; a = area of hole

$$\int_0^v dV = av\int_0^t dt$$

$$\Rightarrow t = \frac{V}{av} = \frac{Ah}{av} = 25$$

30. Tension T = weight of cat (since cat is at rest).

$\therefore$ Net downward force on rod = T + 4g

$$a_{rod} = \frac{3g+4g}{4} = 7g/4\ m/s^2$$

31. Length of the total wire = $l^1 = 3l + \frac{3l}{2} + \frac{3l}{8} ..... = 6l$

$$M = \lambda\cdot(6l) = 100\times 6\times\frac{1}{10} = 60 \text{ grams}$$

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32. Length of the total wire = $l^1 = 3l + \frac{3l}{2} + \frac{3l}{8} ..... = 6l$

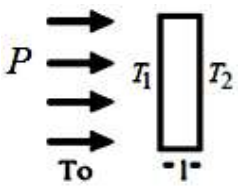
$$M = \lambda\cdot(6l) = 100\times 6\times\frac{1}{10} = 60 \text{ grams.}$$

The moment of inertia of the largest triangle = $I_0 = \frac{1}{12}(3l)\lambda l^2 = \frac{\lambda \ell^3}{4}$

The moment of inertia of the nth triangle $I_n = \frac{\lambda}{4}\left(\frac{\ell}{n}\right)^3$

$$I = \sum I_n = \frac{\lambda l^3}{4}\sum_{n=0}^{\infty}\left(\frac{1}{n}\right)^3 = \frac{\lambda l^3}{4}\times 1.2 = 3\times 10^{-5} kg\cdot x^2$$

33&34



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$$\lambda = 4\sigma e A T_0^3$$

$$P = \lambda(T_1 - T_0) + \frac{KA(T_1 - T_2)}{l} \quad \text{..... (i) (For left surface)}$$

$$\frac{KA(T_1 - T_2)}{l} = \lambda(T_2 - T_0) \quad \text{(For right surface)}$$

When thickness is doubled

$$P = \lambda\left(T_1^1 - T_0\right) + \frac{KA(T_1 - T_2)}{2l} \quad \text{..... (iii)}$$

$$\frac{KA\left(T_1^1 - T_2^1\right)}{2l} = \lambda\left(T_2^1 - T_0\right) \quad \text{..... (iv)}$$

On solving $T_1^1 = \frac{2T_1^2 + T_1T_2 - T_0(3T_1 - T_2) - T_2^2}{2(T_1 - T_0)} = 333.33\,K$

$$T_2^1 = \frac{T_1T_2 + T_0(T_1 - 3T_2) + T_2^2}{2(T_1 - T_0)} = 316.67\,K$$

CHEMISTRY

35. $F - Xe - O - O - \overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}} - F$

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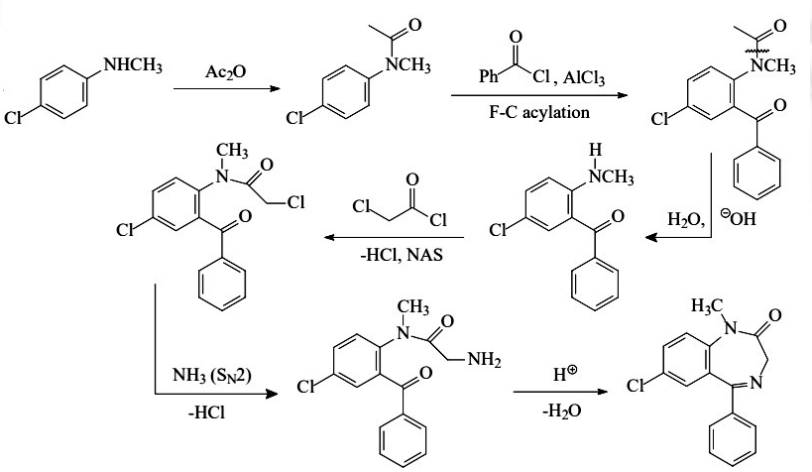
36. Mn^{+2} and Fe^{+3} zero CFSE

37. Since solubility increases on decreasing pH, the anion must be the conjugate base of a weak acid (protonation enhances dissolution).

The constant solubility near neutral pH shows that the cation does not undergo hydrolysis.

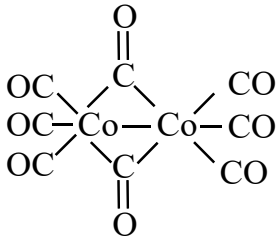
Among the options, only BaF_2 satisfies these conditions.

38.



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39. Oxygen required by microorganisms.

40. 

41. A catalyst is a substance that takes part in a chemical reaction and speeds up the rate but undergoes no permanent chemical change itself. The word catalyst should not be used for any substancethat decreases the rate of a reaction. Catalysts therefore do not appear in the overall balanced chemical equation. But their presence very much affects the rate law, modifying and speeding existing pathways or, more commonly, providing completely new pathways by which a reaction can occur.

42. 0, 1, 2, 3

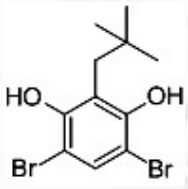
43. $A = SnC\ell_4, E = H_3PO_4$

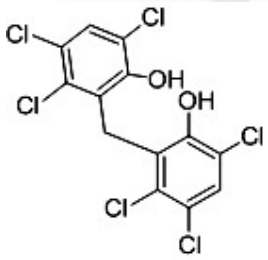
44. $w_{KHP}\text{in 5.8 g soln} = \dfrac{20.4}{118.4}\times 5.8 \Rightarrow n_{KHP} = \dfrac{20.4}{118.4}\times\dfrac{5.8}{204} = n_{NaOH} = n_{AcOH}$

$$\Rightarrow w_{AcOH} = \frac{20.4}{118.4}\times\frac{5.8}{204}\times 60 \Rightarrow \%AcOH = \frac{20.4}{118.4}\times\frac{5.8}{204}\times\frac{60}{5.8}\times 100 = 5$$

45. $$\frac{E2}{E2+E1} = \frac{k_2[tert-butyl\ bromide][HO^-]}{k_2[tert-butyl\ bromide][HO^-]+k_1[tert-butyl\ bromide]}$$

$$\frac{E2}{E2+E1} = \frac{7.1\times10^{-5}\times5.0}{7.1\times10^{-5}\times5.0+1.5\times10^{-5}} = \frac{35.5\times10^{-5}}{35.5\times10^{-5}+1.5\times10^{-5}} = \frac{35.5}{37} = 0.96 = 96\%$$

46. 

47. 

48&49

$$H_2 + I_2 \longrightarrow 2HI$$

$$\Delta H = 435 + 150 - 2\times 300 = -15kJ$$

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initial rate of consumption of $I_2 = -\frac{d[I_2]}{dt} = k[H_2][I_2] = 0.2 \times 0.8 \times 0.4 = 0.064$

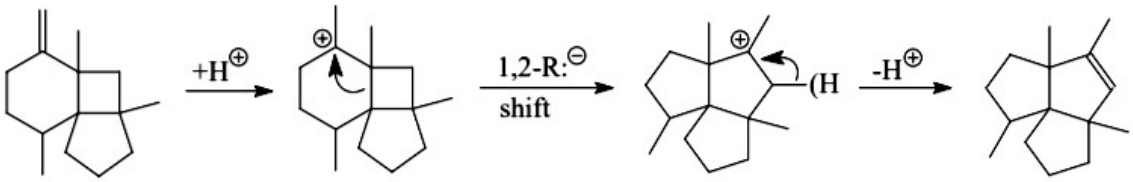
$\therefore$ initial heat liberated $= \frac{15kJ}{1} \times 0.064 = 0.96kJ$

The reaction practically goes for completion at infinite time.

As limiting reagent in I_2, it will be almost completely comsumed

$\therefore$ total heat liberated $= \frac{15kJ}{1} \times 0.4 = 6kJ$

50. There are 5 double bonds across which syn dihydroxylation occurs. In each double bond both OH groups can be either above the plane or below the plane. i.e., across each double bond there are two possibilities of placing the OH groups $\Rightarrow 2^5 = 32$ isomers.

51. 

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